Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 06-Apr-2026

1. Administrative & Study Identification

Full Study Title: A Phase 3b, Open-label Study Evaluating the Efficacy and Safety of Luspatercept (BMS-986346/ACE-536) Initiated at Maximum Approved Dose in LR-MDS With IPSS-R Very Low-, Low-, or Intermediate-risk Who Require RBC Transfusions **Short Title/Acronym:** A Study to Assess Luspatercept in Lower-risk Myelodysplastic Syndrome Participants (MAXILUS) **Protocol Number:** CA056-1060 **NCT Number:** NCT06045689 **Trial Status:** Active, not recruiting **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Luspatercept (BMS-986346 / ACE-536 / Reblozyl) **ATC Code:** B03XA06 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/reblozyl-epar-product-information_en.pdf **Phase of Development:** Phase 3 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy and safety of luspatercept when administered initially at the maximum approved dose (1.75 mg/kg) in lower-risk Myelodysplastic Syndrome (LR-MDS) participants who require red blood cell (RBC) transfusions. * **Secondary Objectives:** To assess the duration of red blood cell transfusion independence (RBC-TI), changes in hemoglobin levels, and the overall safety and tolerability of initiating the drug at its maximum dose.

2.2 Background & Rationale To address the clinical need for optimizing anemia treatment in patients with transfusion-dependent LR-MDS. While luspatercept is an approved erythroid-maturation agent typically initiated at 1.0 mg/kg and titrated upward, this study investigates if starting patients directly on the maximum approved dose of 1.75 mg/kg can more rapidly or effectively induce RBC-TI and improve clinical outcomes for both erythropoiesis-stimulating agent (ESA)-naive and ESA-refractory/relapsed patients without compromising safety.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** N/A (Non-Randomized, Parallel Assignment into ESA-naive and ESA-relapsed/refractory cohorts)

3.2 Planned Enrollment & Duration * **Target N:** 106 participants (Actual Enrollment: 106) * **Number of Sites:** 52 locations globally (including the US, Belgium, Czech Republic, France, Germany, and Italy) * **Estimated Study Start:** 05-Oct-2023 * **Estimated Primary Completion:** 01-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Documented diagnosis of Myelodysplastic Syndrome (MDS) according to World Health Organization (WHO) classification meeting the Revised International Prognostic Scoring System (IPSS-R) classification of very low-, low-, or intermediate-risk disease. 3. Participant has an Eastern Cooperative Oncology Group (ECOG) performance score of 0, 1, or 2. 4. Participant must require red blood cell (RBC) transfusions according to study criteria.

4.2 Exclusion Criteria 1. Known clinically significant anemia due to iron, vitamin B12, or folate deficiencies, autoimmune or hereditary hemolytic anemia, or gastrointestinal bleeding. 2. Prior allogeneic or autologous stem cell transplant. 3. Known history or diagnosis of Acute Myeloid Leukemia (AML). 4. Uncontrolled hypertension.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Luspatercept initiated at the maximum approved dose of 1.75 mg/kg once every 3 weeks. * **Route of Administration:** Subcutaneous injection. * **Duration of Treatment:** Up to 24 weeks for the primary evaluation period, followed by an extension phase and long-term follow-up.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Best supportive care, including RBC transfusions as needed. * **Prohibited:** Disease-modifying agents for MDS and concurrent use of erythropoiesis-stimulating agents (ESAs) during the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, MDS Diagnosis Confirmation, Baseline Labs
Baseline	Day 0	Cohort Assignment, First Dose of Luspatercept (1.75 mg/kg), Vital Signs
Interim	Weeks 1 to 24	Subcutaneous dosing every 3 weeks, Safety Labs, Transfusion Tracking, Hb Assessment
End of Study	Week 24+	Efficacy Assessment (RBC-TI achievement), Long-term Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Number of participants who achieve red blood cell transfusion independence (RBC-TI) for 8 weeks with a simultaneous mean hemoglobin (Hb) increase of ≥ 1 g/dL from Week 1 to Week 24. * **Measurement Method:** Clinical evaluation of transfusion records and central/local laboratory analysis of hemoglobin levels.

7.2 Secondary & Exploratory Endpoints * Number of participants with a mean change in total RBC units transfused over a fixed 16-

week period from Week 9 to Week 24 and from Week 33 to Week 48. * Number of participants who have a time from first dose to first onset of RBC-TI $\ge$ 8-, 12-, and 16-weeks from Week 1 to Week 24, from Week 1 to Week 48, and from Week 1 through EOT. * Number of participants who achieve RBC-TI over any consecutive 8-week period from Week 1 to Week 24, from Week 1 to Week 48, and from Week 1 to EOT. * Achievement of RBC-TI for ≥ 12 weeks. * Number and severity of Treatment-Emergent Adverse Events (TEAEs) to evaluate tolerability. * Duration of response (RBC-TI).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection via site visits, patient diaries, and laboratory assessments (focusing on TEAEs and dose delays/reductions).
- **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness.
- **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Bristol-Myers Squibb to oversee the higher starting dose safety profile.

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy; Safety Set for adverse events.
- **Sample Size Justification:** Approximately 106 participants powered to evaluate the efficacy rate of RBC-TI when bypassing the standard dose-titration schedule.
- **Handling of Missing Data:** Participants with missing data or who discontinue treatment prior to 24 weeks without achieving RBC-TI are generally considered non-responders.

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence and accurate RBC transfusion tracking.
- **Audit Trail:** eCRF locking, tracking of hematology labs, and data clarification forms via standard clinical operations audits.

11. Study Identifiers & Governance

- **Primary ID:** CA056-1060
- **Registry Number:** NCT06045689
- **Sponsor/Lead Organization:** Bristol Myers Squibb
- **Study Title:** MAXILUS
- **Official Regulatory Title:** A Phase 3b, Open-label Study Evaluating the Efficacy and Safety of Luspatercept

(BMS-986346/ACE-536) Initiated at Maximum Approved Dose in LR-MDS With IPSS-R Very Low-, Low-, or Intermediate-risk Who Require RBC Transfusions (MAXILUS)

12. Clinical Framework (Myelodysplastic Syndrome Specific)

- **Primary Condition:** Lower-risk Myelodysplastic Syndromes (LR-MDS) with Anemia
- **Preferred UMLS Name:** Myelodysplastic Syndromes
- **Semantic Type:** Neoplastic Process
- **Diagnosis Threshold:** IPSS-R Very Low-, Low-, or Intermediate-risk disease requiring RBC transfusions
- **Medical Methodology:** Erythroid-maturation agent therapy
- **Optimization Target:** RBC Transfusion Independence (RBC-TI) $ \ge 8$ weeks & Hb increase ≥ 1 g/dL
- **MeSH Code:** D009190
- **HPO Code:** HP:0002863
- **SNOMED ID:** 109995007

13. Study Procedures & Methodology

- **Management Pathway:** Standard tracking of RBC transfusions and continuous hemoglobin monitoring.
- **Education Protocol (HCP):** Training on protocol for initiating luspatercept safely at the maximum 1.75 mg/kg dose.
- **Education Protocol (Patient):** Injection site care, adverse event reporting, and maintaining transfusion logs.
- **Quality Audit Mechanism:** Independent central review of transfusion records and laboratory outcomes.

14. Observation & Follow-Up Schedule

- **Total Duration:** 24 weeks primary evaluation phase, extended to long-term follow-up (up to study completion in Dec 2027)
- **Onsite Visit Cadence:** Every 3 weeks corresponding with the subcutaneous injection schedule
- **Remote Monitoring Frequency:** Every 4 weeks (Q4W)
- **Checklist Review Interval:** Every 12 weeks (Q12W) to assess prolonged RBC-TI

15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY] | | Lead Statistician | [Name] |*** _____ | [DD-MMM-YYYY] |